Customer Shopping Behaviour Analysis

Uncovering patterns in customer spending, preferences, and purchasing behavior across 3,900 transactional records to support smarter business decisions.

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Dataset at a Glance

3,900	18	37
Purchase Records	Features / Columns	Missing Ratings

Key Data Dimensions

- **Customer Info:** Age, Gender, Location, Subscription
- **Product Details:** Category, Size, Color, Season
- **Transactions:** Purchase Amount, Discount Applied
- **Behavior:** Frequency, Review Rating, Shipping Type

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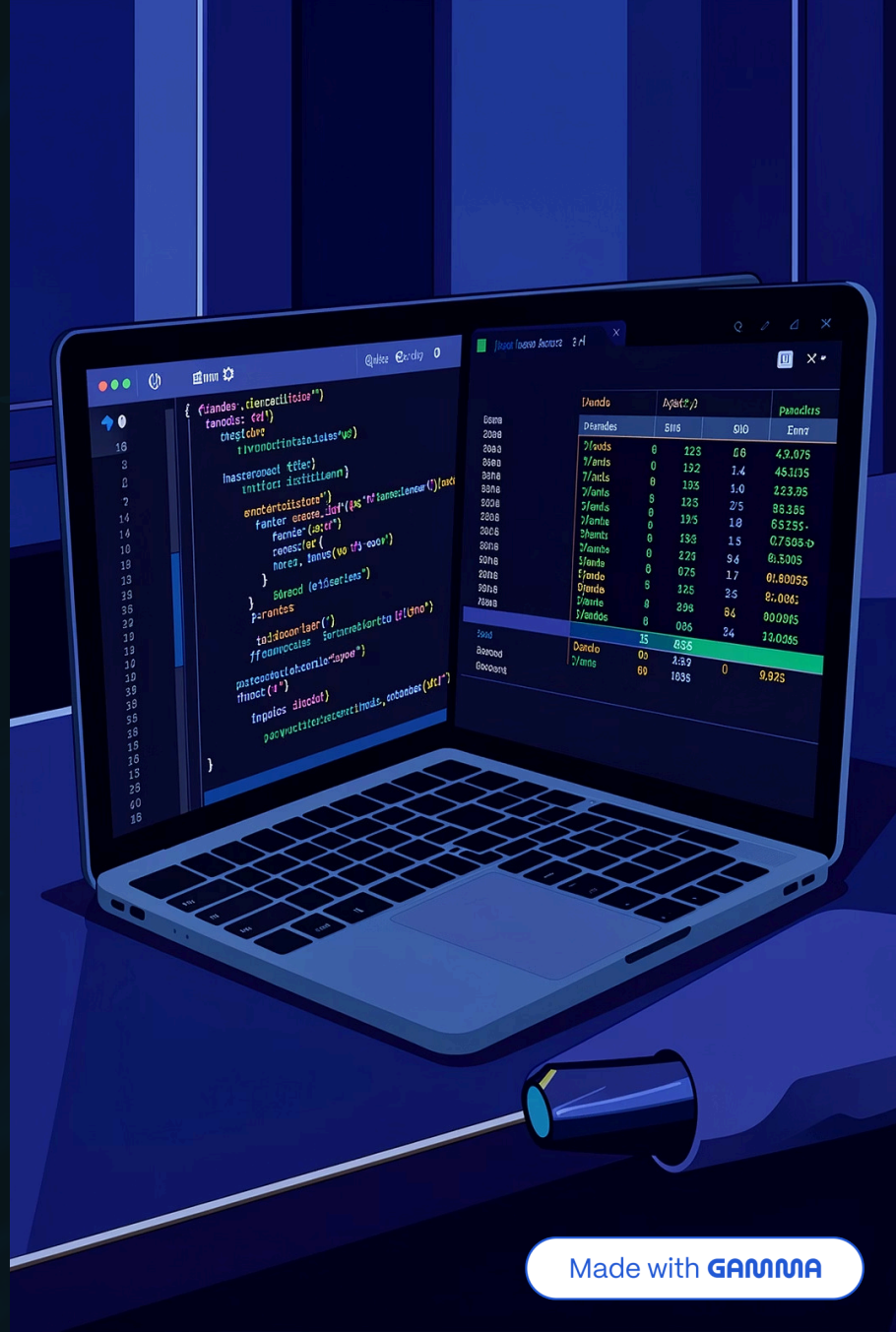
Data Pipeline with Python

The dataset was loaded and processed entirely using **pandas** — no database required.



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Data Preparation Steps

01

Load & Explore

Loaded CSV with pandas; inspected structure via `info()` and `describe()`

02

Clean Missing Data

Filled 37 missing *Review Rating* values using the median per product category

03

Standardize Columns

Converted all column names to `snake_case` for consistency

04

Feature Engineering

Created **age_group** and **customer segments**: New, Returning, and Loyal

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Business Analysis: Key Insights

Revenue by Gender

Compared total spending between male and female customers

High-Spend Discount Users

Customers using discounts who still spent above average

Subscribers vs Non-Subscribers

Spending behavior compared by subscription status

Repeat Buyer Behavior

Frequent buyers more likely to subscribe

Top Rated Products

Products with the highest average review ratings

Revenue by Age Group

Each age segment's contribution to total revenue

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More Key Insights

Shipping Type Analysis

Average purchase value compared across different shipping methods

Discount-Dependent Products

Products most frequently purchased with discounts applied

Top Products by Category

Best-selling items identified within each product category

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Power BI Dashboard

The cleaned dataset was imported into **Power BI** to build an interactive dashboard covering revenue by gender, category-wise sales, customer segmentation, subscription behavior, and shipping trends.

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Customer Segmentation

Segmentation Logic

Customers were grouped using the **number of previous purchases** as the primary signal.

- **New Customers:** First-time or minimal purchase history
- **Returning Customers:** Moderate engagement
- **Loyal Customers:** High frequency, strong retention potential

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Business Recommendations

1

Boost Subscriptions

Offer exclusive benefits to convert more users to subscribers

2

Loyalty Programs

Introduce retention rewards for repeat and loyal customers

3

Optimize Discounts

Balance discount strategy to maintain profitability

4

Promote Top Products

Focus marketing on highest-rated and best-selling items

5

Targeted Marketing

Prioritize high-value segments and key age groups

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Conclusion

This project demonstrates how customer shopping behavior can be effectively analyzed using a CSV-based workflow with Python and Power BI.

Simpler

No database dependency — pandas handles all analysis

Flexible

Easily adaptable to new data or business questions

Insightful

Delivers actionable business value end-to-end

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